

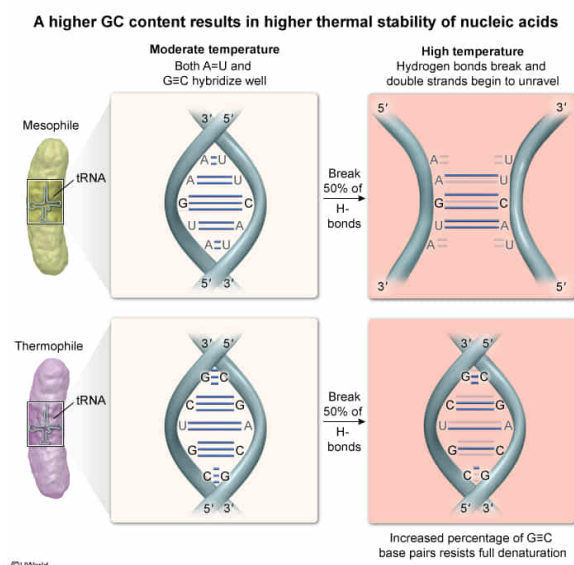
Researchers have found that the nucleotide composition of structural RNAs (eg, tRNA, rRNA, other RNAs that show intra- and inter-strand hybridization) differs between thermophiles, which thrive at elevated temperatures, and mesophiles, which thrive at moderate temperatures. Compared with the structural RNA of mesophiles, the structural RNA of thermophiles most likely contains a higher proportion of:

- ☐ A. adenine and guanine nucleotides. [5%]
- ☐ B. adenine and uracil nucleotides. [5%]
- ☐ C. cytosine and uracil nucleotides. [2%]
- ✔ ☒ D. guanine and cytosine nucleotides. [85%]

Correct

86% answered correctly

Explanation:



Nucleic acid strands can bond non-covalently to a second nucleic acid strand to form a double-stranded duplex called a **double helix**. These non-covalent bonds arise from **purine–pyrimidine base pairs** stabilized via **hydrogen bonds**: guanine (G) base pairs with cytosine (C) via **three hydrogen bonds**, whereas adenine (A) base pairs with either thymine (T) or uracil (U) via two hydrogen bonds. In living cells, DNA strands are most commonly found as part of a double helix with another DNA strand; however, RNA strands can be found both as single strands (eg, mRNA) and as part of a double helix (eg, structural RNAs such as tRNA and rRNA).

For these structural RNAs, the stability of the RNA double helix is critical. Consequently, thermophilic organisms, which live and thrive at **elevated temperatures**, exhibit evolutionary changes to **prevent heat-induced denaturation** of these RNAs (ie, to increase their **melting temperature** T_m). A larger percentage of **G≡C base-pairs**, which involve **more hydrogen bonds** than either A=T or A=U base-pairs, is associated with a **larger, more stable** T_m ; therefore, thermophile structural RNA sequences are more likely to

have a **higher proportion of guanine and cytosine nucleotide residues** to make the RNA duplexes more heat-stable.

(Choice A) Adenine and guanine are the two purines found in nucleic acids. An increasing percentage of adenine (and thus A=T or A=U base-pairs) *lowers* the T_m of nucleic acid duplexes.

(Choice B) A high proportion of A=U base pairs would make an RNA duplex *more* susceptible to heat-induced denaturation and would be less likely in thermophiles, which thrive at elevated temperatures.

(Choice C) Cytosine and uracil are two pyrimidines found in RNA. An increasing percentage of uracil (and thus A=U base-pairs) *lowers* the T_m of nucleic acid duplexes.

Educational objective:

In nucleic acid duplexes, G forms three hydrogen bonds with C, whereas A forms two hydrogen bonds with either T or U. An increasing percentage of G≡C base-pairs increases the thermal stability of the duplex.

Time spent:0 sec

Subject:
Biochemistry

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System:
Carbohydrates, Nucleotides, and Lipids